

REQUIREMENT ANALYSIS PHASE

ComicCraft - AI Comic Story Creator

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Team ID	UNM1401VisionTech
Project Name	ComicCraft - AI Comic Story Creator

1. Customer Journey Map

Map out the customer's experience stage-by-stage.

Stage	Actions	Touchpoint	Customer Thought	Customer Feeling	Process Ownership	Opportunities
Stage 1: Input & Setup	User enters story prompt, selects character, tone, and art style.	Homepage Input Form (Web UI).	"I hope the AI understands my story idea correctly."	Curious and excited.	Frontend/UI Team.	Add a "Random Prompt" button for quick testing.
Stage 2: Generation & Preview	Waits for AI to process and reads the generated story/images panel-by-panel.	Comic Preview Page.	"Wow, these images and dialogues look like a real comic book!"	Amazed and engaged.	Backend & AI Team.	Allow users to regenerate a single panel if they dislike it.
Stage 3: Export & Download	Clicks the "Download as PDF" button.	Export Button & Success Page.	"I want to save this and share it with my friends."	Satisfied and accomplished.	Output Team.	Add direct sharing links for WhatsApp/Instagram.

2. Data Flow Diagram (DFD Details)

Show how data moves between external entities, processes, and data stores.

External Entities: User, Google Gemini API, Stable Diffusion API.

Processes: 1. Process Input, 2. Generate Story, 3. Generate Images, 4. Compile PDF.

Data Stores: Local Images Folder, PDF Exports Folder.

Data Flow Steps:

- User sends Story Prompt to the "Process Input" function.
- System sends the validated prompt to Google Gemini API to fetch Comic Dialogues & Outlines.
- System sends specific scene descriptions to Stable Diffusion API to generate Image Panels.
- APIs return the text and image files to the Local Data Store.
- The "Compile PDF" process merges the text and images into a final document.
- System provides the final PDF download link back to the User.

3. Solution Requirements

Define exactly what your solution will do and how well it will perform.

Functional Requirements (FR)

Category	Requirement
External interfaces	The system must securely connect to Google Gemini API (for text) and Stable Diffusion API (for images).
Transactions processing	The backend must handle asynchronous API calls to prevent the app from freezing during generation.
Reporting / Export	The system must compile generated assets and export a downloadable, formatted PDF document.
Business rules	Prompts must not exceed a specific word count limit to ensure API costs and loading times remain manageable.

Non-Functional Requirements (NFR)

Category	Requirement
Performance & Speed	The complete story and image generation process should take less than 60 seconds.

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Reliability & Availability	The application must handle API timeouts gracefully by showing a user-friendly error message.
Usability & Accessibility	The UI must be highly intuitive, requiring zero technical knowledge for a user to generate a comic.

4. Technology Stack

Identify the programming languages, frameworks, databases, and APIs used.

Layer	Technology	Justification
Frontend / Client-Side	HTML5, CSS3, Jinja2	Easy to build lightweight, responsive forms for user inputs.
Backend / Server-Side	FastAPI & Python	Fast processing and native support for asynchronous API calls.
Database / Data Storage	Local File System	Cost-effective way to temporarily store generated images and PDFs before the user downloads them.
Version Control	GitHub	Seamless collaboration, version tracking, and code sharing among team members.
Third-Party APIs	Google Gemini API & Stable Diffusion API	Core AI intelligence required to dynamically generate text and images based on prompts.